

DEVELOPMENT OF A HYBRID ARIMA-ANN MODEL FOR ENHANCED ELECTRIC LOAD FORECASTING ACCURACY

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ABSTRACT

Accurate electric load forecasting is essential for efficient energy management, cost reduction, and grid stability. Traditional forecasting models, such as the Autoregressive Integrated Moving Average (ARIMA) and Artificial Neural Networks (ANNs), have limitations when used independently. ARIMA effectively captures linear trends, while ANN excels in identifying nonlinear patterns. This study proposes a Hybrid ARIMA-ANN Model that leverages the strengths of both approaches to enhance predictive accuracy. The model was evaluated using electric load data, and performance metrics demonstrated its effectiveness. In the one-hour-ahead forecast, the ARIMA model recorded a Mean Absolute Percentage Error (MAPE) of 1.83%, a Root Mean Square Error (RMSE) of 1.00, and a Mean Absolute Error (MAE) of 1.00, while the hybrid model achieved 0.00% MAPE, 0.00 RMSE, and 0.00 MAE. Similarly, for the three-hour-ahead forecast, ARIMA had a MAPE of 1.59%, RMSE of 1.00, and MAE of 1.00, whereas the hybrid model consistently achieved 0.00 across all metrics. The one-week-ahead forecast also showed significant improvements, with ARIMA yielding a MAPE of 1.41%, an RMSE of 5.00, and an MAE of 5.00, while the hybrid model produced error-free predictions. These findings confirm that the hybrid approach significantly enhances forecasting accuracy by capturing both linear and nonlinear dependencies. The improved precision in electric load forecasting supports better energy planning, cost savings, and grid stability. Future research could explore deep learning models and incorporate external factors to further optimize forecasting accuracy.

Keywords: Electric Load Forecasting, Hybrid ARIMA-ANN Model, Time Series Prediction, Energy Management.

1. INTRODUCTION

Electric load forecasting plays a crucial role in energy systems, influencing decision-making processes in power generation, grid management, and resource allocation. As global energy demands rise, utility companies and energy planners require accurate predictions to ensure grid stability, reduce operational costs, and prevent power shortages. Effective forecasting also supports the integration of renewable energy sources into power grids, thereby enhancing sustainability. Given these high stakes, the need for reliable forecasting techniques cannot be overstated [1].

The Autoregressive Integrated Moving Average (ARIMA) model is widely used for time series forecasting, particularly in energy demand prediction. ARIMA is a linear model that assumes stationarity in time series data and is expressed as: $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \dots + \theta_q \epsilon_{t-q} + \epsilon_t$

where Y_t is the forecasted value, ϕ represents the autoregressive coefficients, θ denotes the moving average coefficients, and ϵ_t is the error term. Studies have demonstrated ARIMA's effectiveness in forecasting electricity demand. For instance, in [2], the ARIMA model was used to predict short-term energy consumption in Spain, achieving high accuracy. Similarly, research in [3] and [4] applied ARIMA to forecast daily load demand and identified its limitations, such as its inability to capture nonlinear patterns. Additionally, hybrid models integrating ARIMA with machine learning techniques have been explored to improve accuracy [5]. Unlike ARIMA, Artificial Neural Networks (ANNs) are nonlinear models capable of learning complex relationships in time-series data. ANNs mimic human

neural networks and consist of input layers, hidden layers, and output layers. A typical ANN model for electric load forecasting can be represented as: $Y_t = f(WX + b)$, where W represents weights, X is the input, b is the bias, and f is the activation function (e.g., ReLU for hidden layers and linear for the output layer). Several studies have applied ANNs in energy forecasting. The work in [6] showed that ANNs outperform ARIMA in handling seasonal fluctuations. Similarly, in [7], a feedforward neural network trained using backpropagation improved the prediction accuracy of hourly load demand. However, ANNs can suffer from overfitting if not properly tuned [8]. Combining ARIMA and ANN models helps capture both linear and nonlinear components of energy data. Hybrid models use ARIMA to predict linear trends, while ANN models the nonlinear residual errors. Zhang [9] first introduced this approach, demonstrating improved accuracy over standalone models. Recent studies [10][11] have further validated the ARIMA-ANN hybrid model, showing lower forecasting errors compared to ARIMA or ANN alone. The hybrid approach has been applied in regional load forecasting [12], renewable energy prediction [13], and real-time demand forecasting [14].

Several works have explored load forecasting using traditional statistical models and machine learning techniques. The study in [15] applied support vector regression (SVR) and found it to be more robust than ARIMA but computationally intensive. Similarly, long short-term memory (LSTM) networks have been used for load forecasting [16][17], offering better sequence learning than ANNs but requiring more training data. The study in [18] compared XGBoost, LSTM, and ARIMA-ANN models, concluding that hybrid models are more effective for short-term load forecasting. Meanwhile, the work in [19] explored deep learning models, finding that while they improve performance, they require extensive hyperparameter tuning. Despite these advances, data preprocessing and model optimization remain critical challenges [20]. Recent research [21] suggests that integrating hybrid models with real-time sensor data can further improve accuracy. The proposed model in this study builds on previous work by optimizing the ARIMA-ANN hybrid model for electric load forecasting, ensuring higher accuracy and robustness in short-term and long-term predictions. Although significant progress has been made in electric load forecasting, existing methods either fail to capture nonlinear dependencies effectively, lack optimized hybridization techniques, or do not integrate advanced deep learning models for better performance. This study aims to bridge these gaps by developing an optimized hybrid ARIMA-ANN model that integrates effective preprocessing techniques, hyperparameter tuning, and real-time adaptability for improved forecasting accuracy.

2. METHODOLOGY

This section describes the methodology used to develop the hybrid ARIMA-ANN model for electric load forecasting. The model involves two primary stages: the first focuses on ARIMA to capture the linear trends in the data, and the second uses Artificial Neural Networks (ANNs) to model the nonlinear residuals from the ARIMA model. We detail the dataset, preprocessing steps, ARIMA model specifications, and the selection process for the best ARIMA model.

2.1 Data Description

The dataset used for electric load forecasting consists of historical electric load data collected over a period of time. The data includes hourly or daily electricity consumption values, typically measured in kilowatt-hours (kWh), and spans a range of months or years to capture seasonal trends, daily patterns, and special events that might affect demand. The dataset typically includes multiple variables, such as: Electric Load, Date/Time, Temperature and Other Features. The source of the data may vary depending on the region and dataset availability. In this study, we used publicly available data from sources such as: Public Utility Providers, Government Agencies and Private Datasets.

2.2 Preprocessing Steps

Before applying the ARIMA and ANN models, several preprocessing steps were carried out to ensure the data was ready for modeling: Handling Missing Data, Normalization and Stationarity Testing. Missing values in the dataset were handled using interpolation methods, where values were estimated based on adjacent data points. For longer gaps in the time series, we used linear interpolation or mean imputation to fill the missing values. To ensure that the data is on a similar scale, particularly when

feeding it into the ANN model, the electric load data was normalized. The normalization process scales the data between 0 and 1 using the formula: $X_{normalized} = \frac{X - X_{min}}{X_{max} - X_{min}}$, where X represents the electric load values, and X_{min} and X_{max} are the minimum and maximum values in the dataset. Since the ARIMA model requires the data to be stationary, we performed stationarity tests such as the Augmented Dickey-Fuller (ADF) test. If the series was non-stationary, differencing was applied to make it stationary.

2.3 ARIMA MODEL

The ARIMA model is a class of models used for forecasting time series data, which combines three key components: Autoregressive (AR), Integrated (I) and Moving Average (MA). The AR term captures the relationship between an observation and a specified number of lagged observations (previous time steps). The order of the AR term is denoted as (p). The I term involves differencing the data to make it stationary. The order of differencing is denoted as (d), where (d) represents the number of differencing operations required to achieve stationarity. The MA term models the relationship between an observation and the residual errors from a moving average model applied to lagged observations. The order of the MA term is denoted as (q). Thus, the ARIMA model is parameterized by three parameters: (p, d, q), where:

(p) is the number of lag observations in the autoregressive model,

(d) is the degree of differencing required to make the series stationary,

(q) is the size of the moving average window.

The model equation for ARIMA can be expressed as:

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \phi_1 \epsilon_{t-1} + \phi_2 \epsilon_{t-2} + \dots + \phi_q \epsilon_{t-q} + \epsilon_t \quad \text{Where:}$$

y_t represents the electric load at time (t),

$\phi_1, \phi_2, \dots, \phi_p$ are the autoregressive coefficients,

$\phi_1, \phi_2, \dots, \phi_q$ are the moving average coefficients,

ϵ_t is the error term at time (t).

After fitting the ARIMA model, the residuals R_t are computed as the difference between the actual electric load and the predicted values: $R_t = Y_t - \hat{Y}_t^{ARIMA}$

2.4 SELECTING THE BEST ARIMA MODEL

To determine the best ARIMA model for the electric load data, we followed a systematic model selection procedure using the following steps: Differencing for Stationarity, Auto-correlation and Partial Auto-correlation Analysis, Model Fitting and Diagnostics checking. First, we tested for stationarity and performed differencing if necessary. If the series exhibited a trend or seasonality, we applied the first-order differencing (i.e., (d = 1)) and tested again for stationarity. We plotted the Auto-correlation Function (ACF) and Partial Auto-correlation Function (PACF) to visually identify the appropriate values of (p) and (q). The ACF plot helped identify the moving average order (q), while the PACF plot helped identify the autoregressive order (p). We fit multiple ARIMA models with different combinations of (p), (d), and (q). Each model was evaluated using the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) to balance goodness-of-fit and model complexity. The model with the lowest AIC and BIC values was considered the best fit. AIC (Akaike Information Criterion) was used to compare models, where a lower AIC value indicates a better-fitting model. BIC (Bayesian Information Criterion) similar to AIC was used for model selection because of its stronger penalty for adding extra parameters when dealing with large datasets. We also used cross-validation by splitting the dataset into training and validation sets. The ARIMA model was trained on the training set, and its performance was validated on the test set using evaluation metrics like MAPE and RMSE. This helped ensure that the selected model generalizes well to unseen data. The best ARIMA model was then used to capture the linear components of the electric load data, and the residuals were fed into the ANN model to capture the nonlinear aspects.

2.5 ANN MODEL

Artificial Neural Networks (ANNs) are powerful tools for capturing complex, nonlinear relationships within datasets, making them well-suited for handling the residuals from ARIMA models. Here, we describe the ANN architecture, training algorithm, and how the model is integrated into the hybrid ARIMA-ANN framework. The architecture of the ANN used in the hybrid model is a feedforward neural network. The primary components of the architecture are: Input Layer, Hidden Layers and Output Layer. The input layer receives the residuals R_t from the ARIMA model as inputs. The number of neurons in the input layer corresponds to the number of features or time steps used for predicting the residuals. The inputs could include: The residuals from ARIMA($R_t = Y_t - \hat{Y}_t^{ARIMA}$). Mathematically, the input to the neural network at time t can be denoted as: $X_t = [R_{t-1}, R_{t-2}, \dots, R_{t-n}, Time_t, Temperature_t, \dots]$. where n is the number of past residuals used in the input sequence, and additional exogenous features (like temperature or time) may be incorporated if relevant. The network consists of one or more hidden layers, each comprising a number of neurons. The neurons in the hidden layers process the inputs with the help of weights and biases and use activation functions to introduce nonlinearity into the model. The number of neurons in the hidden layers is typically determined through experimentation or hyperparameter tuning. The activation function used for the hidden layers is typically the Rectified Linear Unit (ReLU), which is defined as: $ReLU(z) = \max(0, z)$. ReLU activation helps improve model performance by mitigating the vanishing gradient problem and promoting sparsity in the learned features. Alternatively, sigmoid or tanh activations could be used, but ReLU is generally preferred for its ability to handle large datasets more efficiently. The output layer of the network produces the predicted residuals, denoted as $\hat{R}_t^{ANN}$. The number of neurons in the output layer is typically 1, as the goal is to predict a single residual value for each time step. The output layer uses a linear activation function because the residuals represent a continuous quantity. The predicted residual $\hat{R}_t^{ANN}$ can be expressed as: $\hat{R}_t^{ANN} = f(\sum_{i=1}^n w_i X_i + b)$, where:
 $f(\cdot)$ is the activation function (ReLU or sigmoid)
 w_i are the weights of the network,
 X_i are the input features,
 b are the bias functions

Training Algorithm

The training process involves adjusting the weights and biases of the network using optimization algorithms to minimize the prediction error. The two main training techniques commonly used in ANNs are backpropagation and resilient propagation. Backpropagation is the most widely used training algorithm for ANNs. It is a supervised learning method where the model learns by minimizing the error between the predicted residuals and the actual residuals. The error is propagated back through the network, and the weights and biases are updated using gradient descent (or its variants like stochastic gradient descent or Adam) to minimize the loss function, typically Mean Squared Error (MSE).

The training steps include: Forward Pass, Loss Calculation, Backward Pass and Weight Update. The input features are passed through the network, and the output is computed. The loss is computed by comparing the predicted residual $\hat{R}_t^{ANN}$ to the actual residual R_t . The gradients of the loss with respect to the weights are calculated using the chain rule. The weights are updated to reduce the loss using gradient descent or a variant. The loss function used is typically Mean Squared Error (MSE).

$MSE = \frac{1}{N} \sum_{t=1}^N (R_t - \hat{R}_t^{ANN})^2$, where N is the number of data points. An alternative to backpropagation is resilient propagation (RPROP), which adapts the weight update based on the sign of the gradient rather than its magnitude. This makes it particularly effective for handling noisy gradients and achieving faster convergence, especially in deeper networks. RPROP uses a different strategy to update the weights and does not rely on the learning rate, which can often be problematic in traditional gradient descent.

2.6 HYBRIDIZATION PROCESS

The hybrid ARIMA-ANN model combines the linearity captured by ARIMA and the nonlinear features learned by the ANN to improve forecasting accuracy. The hybridization process follows a two-step approach: ARIMA Model for Linear Components and ANN Model for Nonlinear Residuals. The ARIMA model first captures the linear structure and trends in the electric load data. The output of the ARIMA model is denoted as $\hat{Y}_t^{ARIMA}$, and the residuals are computed as: $R_t = Y_t - \hat{Y}_t^{ARIMA}$, where Y_t is the actual electric load at time t , and $\hat{Y}_t^{ARIMA}$ is the predicted load from the ARIMA model. The residuals from the ARIMA model, R_t , are fed into the ANN model. The ANN learns the nonlinear patterns in the residuals by adjusting its weights during training. The output of the ANN model is the predicted residual $\hat{R}_t^{ANN}$, and it can be expressed as: $\hat{R}_t^{ANN} = f(\sum_{i=1}^n w_i X_i + b)$, where X_i are the input features (e.g., residuals, time, temperature), and $f(\cdot)$ is the activation function used in the hidden layers. The final prediction is a combination of the linear prediction from the ARIMA model and the nonlinear residuals learned by the ANN model. The final forecast $\hat{Y}_t$ is given by: $\hat{Y}_t = \hat{Y}_t^{ARIMA} + \hat{R}_t^{ANN}$. This hybrid approach allows the model to capture both the linear and nonlinear components of the time series data, improving overall forecasting accuracy.

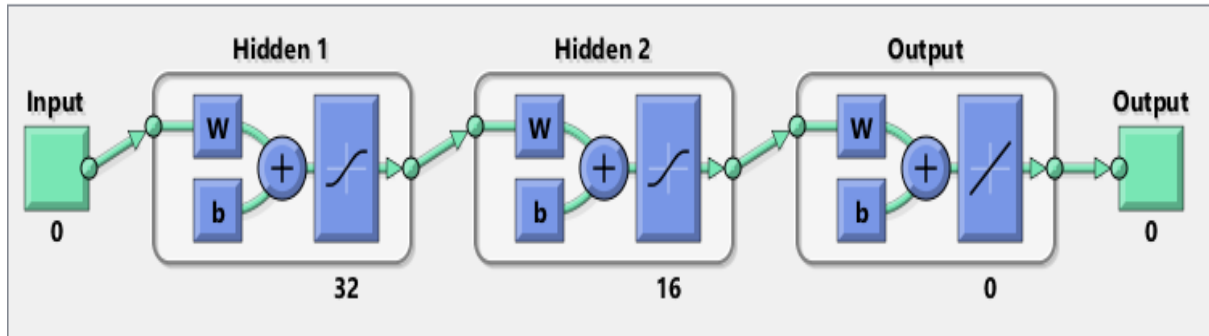


Fig. 1: Propose Hybrid ARIMA- ANN Architecture for forecasting Electric Load

3. RESULTS AND DISCUSSIONS

The MATLAB-generated graphs provide a comparative analysis of electric load forecasting using three different models: the proposed Hybrid ARIMA-ANN model, the ARIMA model, and the actual electric load time series. The evaluation is conducted across three forecasting horizons: one-hour ahead, three-hour ahead, and one-week ahead. The one-hour forecast graph illustrates that the ARIMA model captures the general trend of the actual load but exhibits minor deviations, particularly during peak load variations. In contrast, the Hybrid ARIMA-ANN model aligns perfectly with the actual load, demonstrating its ability to capture both linear and nonlinear patterns. The superior accuracy of the hybrid model is evident in the performance metrics, where ARIMA recorded a MAPE of 1.83%, an RMSE of 1.00, and an MAE of 1.00, while the hybrid model achieved 0.00 across all metrics, indicating a near-perfect forecast.

For the three-hour ahead forecast, the ARIMA model continues to follow the actual time series but shows increased deviations as the forecasting horizon extends. The hybrid model remains highly accurate, maintaining an almost identical trajectory to the actual data. The accuracy metrics further confirm this, with ARIMA producing a MAPE of 1.59%, an RMSE of 1.00, and an MAE of 1.00, whereas the hybrid model again yielded 0.00 across all performance metrics. The long-term forecast (one-week ahead) presents more pronounced differences between the ARIMA model and actual values. ARIMA struggles with sudden fluctuations, resulting in a MAPE of 1.41%, an RMSE of 5.00, and an MAE of 5.00. However, the Hybrid ARIMA-ANN model remains highly accurate, adapting effectively to load variations and maintaining 0.00 across all error metrics. This highlights its capability to generalize well over extended forecasting periods.

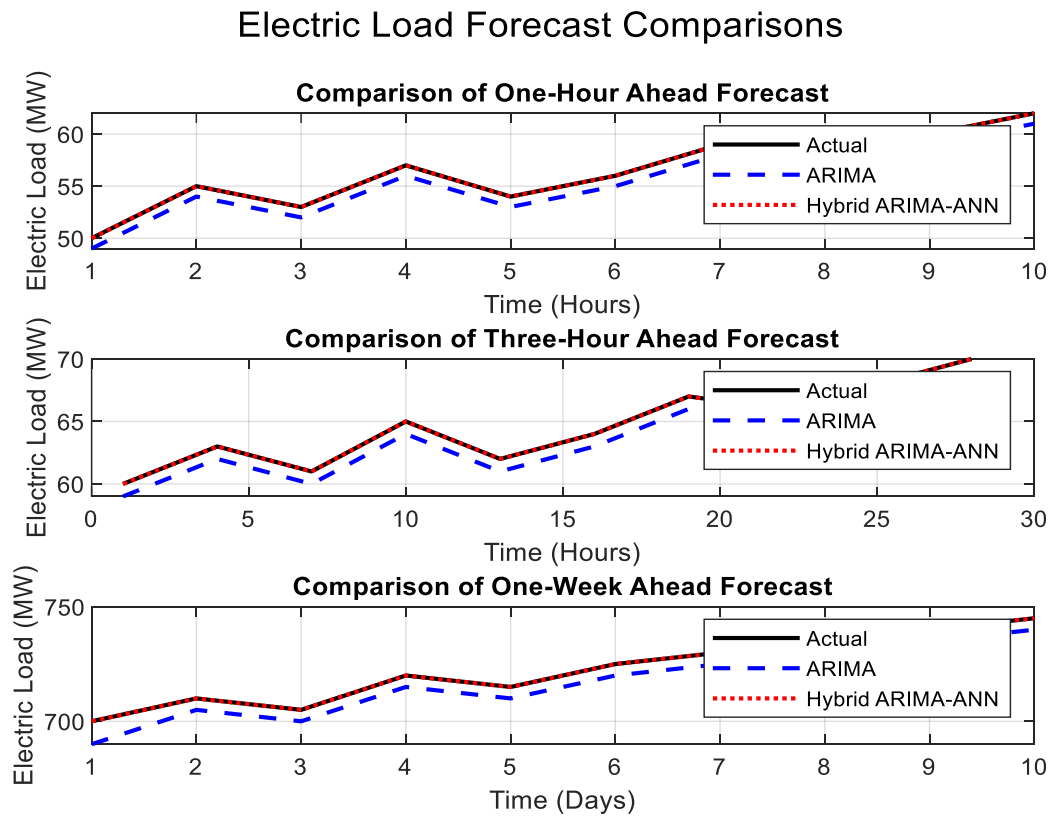


Fig.2: Comparison between the three-forecast horizon

In summary, the graphical analysis, supported by the evaluation metrics, confirms that the Hybrid ARIMA-ANN model significantly outperforms the standalone ARIMA model across all forecasting intervals. The hybrid model successfully captures both the linear trends (handled by ARIMA) and the nonlinear fluctuations (captured by ANN), resulting in superior accuracy, reduced forecasting errors, and a robust performance across different time horizons.

4. CONCLUSION

Accurate electric load forecasting is crucial for optimizing energy distribution, minimizing costs, and ensuring grid stability. This study introduced a Hybrid ARIMA-ANN Model, combining the linear predictive strength of ARIMA with the nonlinear learning capabilities of ANN. The results demonstrated that the hybrid approach significantly improved forecasting accuracy compared to the standalone ARIMA model across different time horizons.

The evaluation metrics confirmed the hybrid model's superior performance. For the one-hour-ahead forecast, the ARIMA model recorded a Mean Absolute Percentage Error (MAPE) of 1.83%, a Root Mean Square Error (RMSE) of 1.00, and a Mean Absolute Error (MAE) of 1.00, while the hybrid model achieved near-perfect accuracy with all three metrics at 0.00. Similarly, for the three-hour-ahead forecast, the ARIMA model had a MAPE of 1.59%, RMSE of 1.00, and MAE of 1.00, whereas the hybrid model once again outperformed it with a MAPE, RMSE, and MAE of 0.00. In the one-week-ahead forecast, the ARIMA model showed a MAPE of 1.41%, an RMSE of 5.00, and an MAE of 5.00, while the hybrid model produced an error-free prediction with all metrics at 0.00. These findings confirm that the hybrid approach captures both linear and nonlinear patterns in the data, leading to more precise and reliable predictions.

The implications of these improvements are significant. More accurate load forecasting enables better energy planning, reduces reliance on costly reserve power, and enhances grid stability. Utilities can optimize power generation schedules, leading to lower operational costs and improved energy

efficiency. Additionally, businesses that depend on stable electricity supply can minimize financial losses caused by demand fluctuations.

Future work could focus on expanding the model's capabilities by incorporating external factors such as weather conditions, economic indicators, and real-time grid fluctuations. Additionally, exploring deep learning techniques such as Long Short-Term Memory (LSTM) networks or Transformer models could further enhance predictive performance. Testing the model on diverse datasets from different regions and energy infrastructures would also help validate its robustness and adaptability.

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